

Limit of Densifying WiFi

Yuxuan Sun, Chenyu Xu, Haojie Hu

Abstract—The increasing demand for high-speed and reliable WiFi connectivity necessitates efficient access point (AP) deployment strategies. This study explores the relationship between AP density and network performance, particularly focusing on throughput and Signal-to-Interference-plus-Noise Ratio (SINR) in dense WiFi deployments. Using simulation-based methods, we analyze the impact of interference management, with a focus on the Simple Sequential Inhibition (SSI) mechanism, on achieving optimal network performance. Results demonstrate that while initial increases in AP density improve throughput, excessive density leads to saturation due to interference. The SSI mechanism effectively mitigates interference, stabilizing performance and enabling efficient spatial reuse. These findings highlight the importance of balancing AP density and interference management in optimizing WiFi networks.

Index Terms—AP, WiFi, AP density, AP coverage, throughput, SINR

I. BACKGROUND

A. Why Important

There are many evidences showing that the problem about WiFi performance and AP density means a lot:

The rapid expansion of wireless communication networks and the growing reliance on WiFi for connectivity in both urban and rural areas make the optimization of access point (AP) deployment a critical issue. As the demand for high-speed, reliable, and low-latency wireless networks continues to surge, understanding the trade-offs between AP density, interference management, and network performance has become more urgent than ever.

A poorly optimized AP deployment not only results in inefficient use of limited frequency resources but also degrades user experience through increased interference and signal degradation. This has implications for a wide range of applications, from residential WiFi performance to industrial IoT systems and smart cities. Identifying the optimal AP density and coverage strategies can significantly enhance network efficiency, reduce energy consumption, and enable seamless connectivity in high-demand environments.

B. Goals

Our research goals are:

Quantify the relationship between WiFi APs density and area throughput

Identify the limits of throughput: attainable throughput in dense deployments

Evaluate strategies: mitigate interference and improve throughput. e.g. planned frequency allocation

C. Hypothesis

It is hypothesized that the relationship between AP density and network performance follows a non-linear trend. Specifically:

Low AP density results in poor signal coverage, as users are far from APs, causing weak signal strength.

Moderate AP density achieves the optimal balance between throughput and Signal-to-Interference-plus-Noise Ratio (SINR). High AP density introduces excessive interference, lowering SINR and throughput.

The key to mitigating these issues lies in identifying the optimal AP deployment density that maximizes throughput while minimizing interference.

D. Related Work

Several models and methodologies have been developed to study this problem:

Signal Propagation Models: Analyze signal attenuation based on distance, such as the Free Path Loss Model.

Interference Models: Quantify the effects of overlapping signals and their impact on SINR.

Heatmap Simulation Models: Visualize SINR distributions and identify regions of poor signal coverage or high interference.

Monte Carlo Simulations: Provide insights into the trade-offs between AP density, coverage radius, and user performance in stochastic environments.

These studies highlight the challenges of dense WiFi deployments and emphasize the need for adaptive strategies to optimize network performance.

E. Experimental Problems

While these models provide a strong foundation, several practical challenges remain:

Channel Modeling: Current models may oversimplify channel conditions, failing to capture real-world complexities.

AP Coverage Overlap: Randomized AP placement may lead to overlapping coverage areas, further intensifying interference.

Parameter Realism: Setting realistic parameters (e.g. frequency-band allocation, coverage radius) is critical to ensure simulation results are applicable to real-world scenarios.

Dynamic Adjustments: The relationship between AP range and frequency must be systematically integrated into the model to reflect practical deployment constraints.

By addressing these gaps, the study aims to refine existing models and propose a robust framework for optimizing WiFi network deployments in diverse environments.

II. METHOD

A. Scenario Implementation:

1) *Simulation Environment*: The aim of this study is to investigate the user throughput performance of 5GHz band WiFi networks under different AP density deployment conditions.

The simulation environment is set as a space of limited dimensions without considering other fading. All APs are considered as line-of-sight (LoS) propagation to the users. In this environment, only the free space propagation loss is considered for the path loss, and its specific calculation method will be described in detail in the Methods section. The simulation area is set as a rectangular area of 5km5km in which APs are deployed in uniform network format and users are randomly distributed within the area.

2) *AP Deployment Method*: WiFi is deployed in a grid to ensure uniform coverage over the entire simulation area.

AP deployment density λ_a (in AP/m²) is used as the key variable with a range of values expressed as, e.g. $\lambda_a \in [0.00001, 0.000015]$

For a given λ_a , the APs are evenly spaced to optimize spatial reuse capability.

All APs operate in the 5GHz band, and the transmit power of each AP is set to 20 dBm. 24 non-overlapping channels are available to the system using the 802.11ac standard and assuming that all APs use a 20MHz channel width. The specific channel allocation strategy is further described in the Methodology section.

Table 1: Wi-Fi channel division when operating in the 5 GHz band¹²

Frequency Band ¹³	Usable 20MHz Channel Numbers ¹⁴	Number of Available Channels ¹⁵
UNII-1(5.150-5.250GHz) ¹⁶	36, 40, 44, 48 ¹⁷	4 ¹⁸
UNII-2A(5.250-5.350GHz) ¹⁹	52, 56, 60, 64 ²⁰	4 ²¹
UNII-2C(5.470-5.725GHz) ²²	100-144 ²³	11 ²⁴
UNII-3(5.725-5.850GHz) ²⁵	149, 153, 157, 161, 165 ²⁶	5 ²⁷
Total ²⁸	36-165 ²⁹	24 ³⁰

3) *User Distribution*: User devices are randomly and uniformly distributed in the simulation area.

The number of users in each simulation experiment is customisable to simulate user access under different AP densities. However, to control the variables, we set the number of users to a fixed value in our experiments.

The minimum reception sensitivity of the user's device is set to -85 dBm, which means that the user needs to receive a signal of at least -85 dBm in order to successfully connect to a particular AP. the user always connects to the AP that provides the strongest signal to ensure the best access experience.

4) *Carrier Listening and Interference Modelling*: WiFi networks use the Carrier Sensing Multiple Access/Conflict Avoidance (CSMA/CA) mechanism for transmission control. In this mechanism, the AP needs to perform Carrier Sensing to determine if parallel transmission is possible.

In this study, the Carrier Sensing Threshold ($CS_{threshold}$) of the APs is set to -82 dBm to determine the presence of

interfering signals. When the received power between APs exceeds this threshold, they back off to avoid conflicts.

III. MODEL

A. System Modelling

General framework for modelling:

- Wireless channel modelling: calculating WiFi signal propagation path loss. free space path loss.
- AP deployment: generates AP locations in the grid based on AP density.
- SSI mechanism: controls the AP activation process to avoid channel conflicts.
- User association: selects an AP based on signal strength.
- SINR calculation: analyses the interference between APs.
- Throughput Calculation: Calculates user throughput based on SINR and normalises it to throughput density.

B. Wireless Channel Modelling

Free space path loss (FSPL):

WiFi signals in the 5GHz band are mainly affected by free-space path loss and channel modelling is done using function:

$$L(d_{ij}) = 46.4 + 20\log_{10}(d_{ij}) + 20\log_{10}\left(\frac{f_c}{5}\right)$$

Where $L(d_{ij})$ is the path loss (dB), d_{ij} is the distance from the AP to the user, and f_c is 5 GHz. 46.4 is a constant term that is usually used for benchmarking calculations in the 5 GHz band. $20\log_{10}(d_{ij})$ denotes the path loss versus distance d , consistent with the Friis propagation model. $20\log_{10}(f_c/5)$ denotes the frequency-dependent loss.

In this section, we have not considered other possible attenuations and have only considered free-space path loss. However, in real scenarios, depending on the location of the scene (e.g. office, street, suburb, etc.), there may be different attenuations that need to be taken into account. This will be elaborated in future works.

C. AP Deployment Modelling

Problems with

Traditional random deployment: Random deployment of AP locations may result in uneven distribution among APs and overcrowding in certain areas, causing excessive interference. It may also result in no AP coverage in some areas, causing uneven network coverage.

Gridded AP Deployment: Within a fixed area (5km x 5km), the grid size was calculated based on AP density. Place the AP at regular intervals on the X-Y axis. The role of grid deployment: ensure that APs cover the whole area evenly, reducing the impact of distance variation between APs on throughput. Avoiding overly dense or overly sparse APs helps to more fairly evaluate WiFi performance at different densities.

D. Coverage Range

Coverage determines the maximum distance from which an AP can effectively cover and provide services.

It is calculated based on the FSPL model and depends on the transmission power of the AP and the minimum receive power threshold of the user equipment. During the user association process, the user equipment can only connect to the APs within its coverage area, which directly affects the AP activation process and network load distribution.

In addition, during the user distribution process, the code filters the APs that can provide services based on the coverage area, and selects the optimal AP based on the signal strength, which affects the user's signal quality and network performance.

Coverage is defined as the distance when the received signal power is equal to the minimum receivable power of the user:

$$P_r = P_t - L(d_{coverage})$$

Let $P_r = P_{threshold}$, and P_t be the transmit power of the AP, e.g. 20dBm, this value is the typical transmit power of 802.11 WiFi devices.

From this, the coverage can be solved:

$$d_{coverage} = 10^{\left(\frac{P_t - P_{threshold} - 46.4}{20}\right)}$$

Where $P_{threshold}$ is set to -85dBm, which meets the minimum receive sensitivity for the 5GHz band in the IEEE802.11 standard.

1) *Carrier sensing range*:: The carrier sensing range is used to determine the mutual interference relationship between APs.

If the signal of an AP can still be detected within the carrier sensing threshold of another AP, the AP may be marked as a 'grey AP', indicating that it is interfered with by the signal of a neighbouring AP and does not have the ability to independently schedule a channel.

When calculating the signal-to-interference-and-noise ratio (SINR), only APs within the carrier sensing range that use the same channel as the source of interference are considered, so as to assess the actual communication quality.

The arithmetic of the carrier sensing range is:

$$L(d_{cs})P_t = CS_{threshold}$$

Substitute the path loss model:

$$P_t - L(d_{cs}) = CS_{threshold}$$

Available after collation:

$$d_{cs} = 10^{\left(\frac{P_t - CS_{threshold} - 46.4}{20}\right)}$$

d_{cs} is the carrier sensing range and $CS_{threshold}$ is the carrier listening threshold, which is the minimum detectable signal power when the AP listens to the channel.

The typical value of -82dBm means that if the AP listens to a signal with a strength greater than this value, then the ap must wait and cannot send data; on the contrary, it can send data freely.

E. SSI Mechanism

1) *Principles of the SSI mechanism*:: Here are the theoretical Simple Sequential Inhibition (SSI) mechanism:

(1) Select the first AP.

Must be an AP with at least one user in the coverage area.

(2) Control interference between APs.

When an AP is selected, APs within its carrier sensing range (d_{cs}) are temporarily blocked to prevent simultaneous activation from causing interference.

(3) Select the activatable APs in turn.

Gradually, an AP is randomly selected from the remaining APs to be activated, and the activated AP also blocks the APs within its d_{cs} range.

(4) Until there are no optional APs.

Until all APs are marked as activated or blocked, AP selection ends.

2) *The actual realisation process*:: Steps:

After selecting the initial AP (i.e., the one with users present around it), selecting away from him as the $AP_{serving}$ will deposit the $AP_{serving}$ into the AP_{active} list, which is currently the only active ap. the $AP_{serving}$ is within the D_{cs} range of the $AP_{serving}$, and other APs within this D_{cs} range (carrier sensing range) are set to be temporarily disabled. You should deposit these disabled APs into the grey AP list.

Next, among the APs other than the grey AP and the $AP_{serving}$, randomly select an AP to be the second AP_{active} . for the second AP_{active} , check its D_{cs} range, and for the other aps in this range, make them temporarily disabled and add them to the grey ap list.

Now there are two active ap locations stored in AP_{active} . Among the APs other than these two activated aps and the grey ap, randomly pick a third AP_{active} , repeat the above deactivation of APs, and keep looping until you can't pick any new APs that can be activated, jump out of the loop (at the moment of jumping out of the loop, all the APs have been divided into the two categories of activated and grey APs, and none of the other APs exists), and record the following current AP_{active} list and grey AP list.

It is worth noting that activating APs without associating them with users is meaningless. To activate an AP, you must select a user and the initial AP associated with it (that is, this user's $AP_{serving}$).

3) *Advantage of the SSI mechanism*: Reduces conflicts between APs and improves WiFi spectrum utilisation.

F. User association modelling

User Generation uses a uniform random distribution to generate user locations within a predefined simulation area. This method can simulate the random movement and distribution characteristics of users in real-life scenarios.

During the subsequent AP activation and user association process, user location data is used to calculate the Euclidean distance between the user and the AP to determine whether the user is within the coverage area of an AP.

In addition, the user location is also used to calculate signal strength and SINR to evaluate network throughput and interference under different AP densities.

Steps for users to select the best AP:

a. Calculate the distance from the user to all APs

First traverses all APs and calculates the Euclidean distance between the user and each AP. The Euclidean distance follows the function:

$$d = \sqrt{(x_{user} - x_{AP})^2 + (y_{user} - y_{AP})^2}$$

(x_{user}, y_{user}) is the position of the user, and (x_{AP}, y_{AP}) is the position of AP.

b. Filter APs within coverage

Only APs within the coverage considered optional APs.

c. Calculating the received signal strength

For all candidate APs, the code calculates the received signal strength using the free-space path loss model:

$$P_{rx} = P_t - L(d_{ij})$$

d. Select the AP with the strongest signal.

The user selects the AP with the strongest signal strength from the candidate APs for association. If no AP meets the coverage requirements, the user is not assigned to any AP.

G. SINR calculation

SINR is mainly used to evaluate the signal quality received by the user.

1) *Co-channel Interference*: In a WiFi network, different APs may operate on the same channel, resulting in co-channel interference. When multiple APs are assigned to the same channel, and their signals can be heard by other APs within a certain range, the signal quality of users may be affected. Therefore, in this simulation, we use a channel assignment mechanism to reduce the impact of co-channel interference.

There are usually 24 available channels in a WiFi network (calculated using a 20 MHz channel width), and in order to avoid the severe interference caused by all APs sharing the same channel, APs are randomly assigned to different channels. When calculating the SINR, only APs connected to the user on the same channel are considered as interference sources, while APs on different channels are considered as not directly interfering.

Since WiFi uses the CSMA/CA (Carrier Sense Multiple Access/Collision Avoidance) mechanism, APs within the carrier sense range may affect the SINR calculation. As the density of APs increases, the number of co-channel interference APs rises, resulting in an increase in $P_{interference}$ and a decrease in SINR. If the density of APs is too high, multiple APs may compete for channel resources at the same time, which affects throughput.

In addition, the performance of SINR is also affected by the channel reuse strategy. When there are more available channels, the channel reuse rate between different APs is low, interference is low, and the user's SINR is maintained at a high level. However, when the AP density is too high, channel resources are limited, and the channel reuse rate is increased, co-channel interference becomes more pronounced, affecting the user's communication quality.

2) *SINR calculation*: Interference from other APs:

$$P_{interference} = \sum_{j \neq s, j \in C} (P_t - L(d_{ij}))$$

P_t is the transmission power of the AP, C is the set of all APs using the same channel, and s is the serving AP of user i .

Signal power calculation:

$$P_{signal} = P_t - L(d_{ij})$$

The final SINR formula is obtained as:

$$SINR = 10 \log_{10} \left(\frac{10^{\left(\frac{P_{signal}}{10}\right)}}{10^{\left(\frac{P_{interference}}{10}\right)} + 10^{\left(\frac{P_{noise}}{10}\right)}} \right)$$

P_{signal} is the power of the signal received from the AP selected by the user, $P_{interference}$ is the interference power from other APs, and P_{noise} is the ambient noise power, which is usually set to -100dBm.

H. Throughput calculation

1) *Throughput calculation*: Shannon's formula was used:

$$R = B \log_2(1 + SINR)$$

B is the channel bandwidth (20MHz is used in this paper). According to the 20MHz channel bandwidth in 5GHz WiFi, it is usually considered to be composed of 24 available channels and retrograde multiplexing in practical applications.

I. Monte Carlo simulation

This experiment uses Monte Carlo simulation to evaluate the throughput performance of WiFi networks with different AP densities.

The experiment is carried out in a fixed $5\text{km} \times 5\text{km}$ area, with APs deployed in a grid, and the SSI mechanism is used to select active APs, while channels are assigned to APs to reduce co-channel interference.

At each AP density, users are generated randomly, and the SINR of all users is calculated, considering only the interference of co-channel APs within the carrier sense range. The throughput is calculated based on Shannon's formula to measure the impact of AP density on the overall network throughput and user experience.

The experiment is simulated 10 times independently for each AP density, and the average value is taken to reduce random errors. The curve of throughput versus AP density is plotted to analyze the impact of the throughput saturation point and channel multiplexing strategy on network performance.

IV. RESULTS AND ANALYSIS

We made many mistakes in the previous research before the Final Seminar and we quickly changed our logic and carry on the new experiments:

The new experiments focus on evaluating the impact of WiFi Access Point (AP) deployment density on communication performance, with particular emphasis on throughput and SINR in the fixed frequency. The simulation highlights the influence of density and the role of the SSI mechanism in mitigating interference.

The results illustrate two key phases as AP density increases: an initial growth phase with increasing throughput, and a saturation phase where throughput stabilizes due to interference control enabled by SSI.

A. Previous Research Mistakes

There are lots of mistakes in the old results of previous research:

We focused on the AP density coverage but ignored that there are corresponding models to the AP coverage in real life. Most importantly, we didn't put the frequency at a fixed value but a variable. We correct this mistake in the latter research. And the outcomes showed some errors, e.g. the trend of throughput should not be to increase first and then continue to decrease.

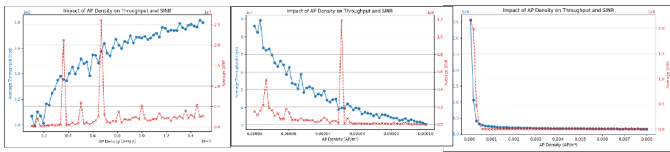


Fig. 1. Throughput and SINR with AP Density of Low, Medium and High

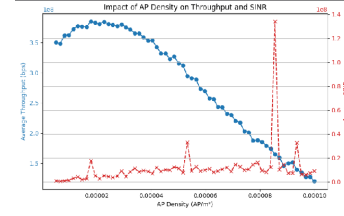


Fig. 2. Throughput and SINR with Ideal AP Density

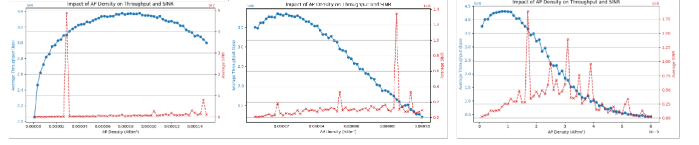


Fig. 3. Throughput and SINR with AP coverage of 30-60m, 50-120m and 80-200m

B. Observed Throughput Growth and Saturation

In the experiments, the mean area throughput density exhibited a distinct trend.

When AP density was low, increasing density led to a significant rise in throughput. This trend aligns with the expectation that additional APs improve spatial reuse and provide more users with access to high-quality connections. However, as density continued to increase, throughput reached a saturation point and no longer improved significantly.

This stabilization occurred because the SSI mechanism selectively activated APs while deactivating others (grey APs) to control interference. As a result, the network avoided the unbounded co-channel interference that typically occurs in high-density deployments without interference management. The control of interference prevented a sharp decline in SINR, ensuring a balance between AP availability and interference suppression.

C. Comparison with previous model

The observed results align partially with prior studies but introduce unique insights due to the role of SSI. In conventional models without interference control, throughput may decline after reaching a peak due to severe interference from active APs sharing the same channel. However, in this simulation, the SSI mechanism ensured that saturation was reached without a decline. This outcome demonstrates the efficacy of SSI in enhancing network stability and capacity.

D. Analysis of Key Factors

1) *SINR*: SINR is a critical metric in determining throughput.

In the initial phase, when AP density was low, SINR remained high due to minimal co-channel interference. This high SINR allowed users to achieve near-maximum data rates. As density increased, interference power also increased, leading to a gradual decline in SINR. However, the SSI mechanism mitigated this decline by limiting the number of active APs within a carrier-sensing range, ensuring SINR remained within a usable range.

2) *Role of SSI Mechanism:* The Simple Sequential Inhibition (SSI) mechanism played a pivotal role in achieving saturation without throughput degradation.

By enforcing spatial reuse constraints, SSI deactivated APs that would cause excessive interference to nearby active APs. This mechanism effectively managed co-channel interference while maintaining a high density of active APs. As a result, the network utilized spatial resources efficiently without compromising SINR. SSI's impact is particularly evident in the stabilization phase. As AP density reached high levels, the number of deactivated (grey) APs increased proportionally. This ensured that the interference footprint of the network remained controlled, preventing a collapse in SINR that could reduce throughput.

3) *Channel Allocation and Interference:* Channel allocation significantly influenced interference patterns in the network.

With a limited number of channels (24 in our simulation), channel reuse became necessary as density increased. Co-channel interference arose when multiple APs within proximity used the same channel. However, SSI reduced the probability of co-channel interference by controlling the activation of APs. Additionally, random channel allocation ensured that interference was distributed uniformly across the network, avoiding localized hotspots of severe interference.

4) *Density, Saturation, and Spatial Reuse:* Spatial reuse, a key advantage of dense deployments, enabled the network to support high throughput in the initial growth phase.

As density increased, more users could connect to nearby APs with strong signal strength, reducing path loss and improving link quality. However, spatial reuse reached a limit as interference became the dominant factor. SSI's role in regulating active APs ensured that this limit was not exceeded, allowing the network to achieve stable performance in the saturation phase.

E. Comparison of Saturation Behavior

1) *Without Interference Management:* In scenarios without interference management, high-density deployments often lead to a sharp decline in throughput after reaching a peak. This decline occurs because the cumulative interference from densely packed APs overwhelms signal power, reducing SINR to unusable levels. Such behavior has been observed in previous studies that lack mechanisms like SSI.

2) *With SSI Mechanism:* New results demonstrate that SSI prevents this decline by maintaining a controlled interference environment.

The saturation phase observed in our experiments represents a state of equilibrium where the benefits of spatial reuse are maximized without excessive interference. This contrasts with unmanaged networks, where saturation is accompanied by a performance collapse.

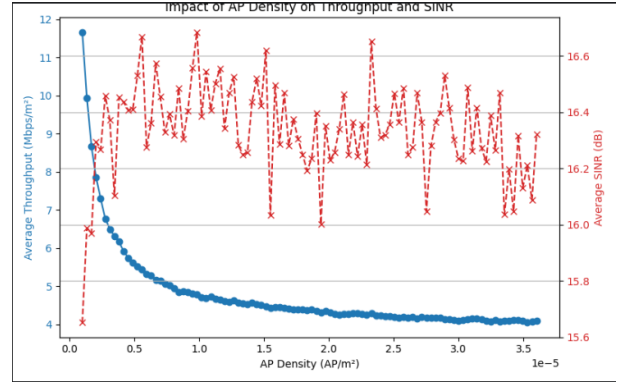
F. Practical Insights

The results provide several practical insights for WiFi network design:

Deploying APs at densities just below the saturation point ensures maximum throughput while minimizing interference. The saturation point varies with environmental factors and interference control mechanisms.

Mechanisms like SSI are essential for achieving stable performance in dense deployments. Networks without such mechanisms risk severe degradation in high-density scenarios.

Efficient channel allocation can further enhance network performance by reducing co-channel interference. Combining SSI with dynamic channel assignment could yield even better results.



1. A very large area and a very small number of users are used, and the density setting is relatively low.

2. The throughput formula is wrong, causing the results to show the trend of average single AP throughput.

V. CONCLUSION

Our experiments demonstrate that:

WiFi network performance in dense deployments depends critically on interference management mechanisms like SSI. While increasing AP density initially boosts throughput, saturation occurs when interference becomes a limiting factor.

SSI ensures that saturation does not lead to performance degradation by controlling active APs and maintaining acceptable SINR levels. These findings highlight the importance of balancing spatial reuse and interference management in dense WiFi networks.

VI. FUTURE WORKS

A. Previous Mistakes

There are 3 main errors in the former research:

The throughput trend should not continuously decline after reaching its peak but should remain at the highest point after saturation, or slightly decline. This could be due to an error in the calculation method of throughput in the code, where total throughput and per-user throughput might have been confused.

Setting frequency as a variable may affect the final results, which is incorrect. Therefore, frequency has now been fixed.

AP coverage in real-world scenarios follows specific models, rather than randomly varying the coverage radius by assigning a range.

B. Path Loss Model

In our project, we only considered free space path loss.

However, in real life, there may be various kinds of losses, such as Multipath fading, Material Absorption, Adjacent Channel Interference and so on.

Later on, the corresponding attenuation can be selected according to the specific scene for the experiment. For example, in a complex office space, the multipath effect and material absorption can be taken into account.

C. AP activation

In our project, our AP activation logic has the following problems:

Even if the first AP is guaranteed to be activated, there is still a chance that a user may not fall within the AP's coverage area during subsequent cycles, which can lead to wasted APs.

There may be cases of multiplexing. For example, when there are two or more users in a coverage area and these users do not have a closer (more powerful) AP to choose from, these users may not be able to connect to the same AP.

D. Future Design

In our project, Shannon's formula is used to estimate the achievable throughput.

However, our formula is actually just an idealised model that lacks the portrayal of the display environment. In this case, the PHY layer factor affects the theoretical maximum throughput and the MAC layer factor affects the actual throughput.

Therefore, in future work, we can consider adding the influence of PHY layer and MAC layer factors to make the experiment closer to the reality.

APPENDIX

Yuxuan Sun:

Report: method, scene implementation and future works section.

Seminar: Seminar 1, 2, 3: Introduction to the method section
Final seminar: Introduction to the analysis of the Results section

Code compilation: Before the final seminar: add code details after Chenyu Xu completes the overall framework.
After the final seminar: sort out and clarify the overall logic of the code, compile the AP activation mechanism in the SSI mechanism, the user selection mechanism, the coverage and carrier listening range compilation, SINR and throughput calculations, and debugging

Chenyu Xu: Report: results

Seminar: Seminar 1,2,3: method and experimental design

Final seminar: Experimental method and process

Code compilation: the overall framework of the code, calculation of SINR and throughput, visualization of result
After the experiment: analyzing the graphs

Haojie Hu:

Report: Background, related work and the whole writing.

Seminar: Seminar 1, 2, 3, final seminar: Introduction to the background, related work conclusion and future works.

Code compilation:

Code of Channel Simulation, running the codes for outcomes.

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